Lesson 10: Modes of Convergence and Borel–Cantelli

AI/ML Mathematics

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1 Motivation and Overview

Every convergence theorem in reinforcement learning—from the basic consistency of Monte Carlo estimation to the asymptotic correctness of TD(λ) and Q-learning—asserts that a sequence of random variables approaches a limit in one of several precisely defined senses. These senses are *not* interchangeable: almost sure convergence, convergence in probability, L^p convergence, and convergence in distribution each capture a different aspect of how random approximations stabilize.

This lesson develops four modes of convergence for sequences of random variables, establishes the complete hierarchy of implications among them, and then proves the Borel–Cantelli lemmas—the primary tool for deducing almost sure convergence from summability of probabilities.

Why This Matters for RL

- **Almost sure convergence** is the gold standard in stochastic approximation: the TD(0) convergence theorem establishes $V_t(s) \xrightarrow{\text{a.s.}} V^\pi(s)$.
- **Convergence in probability** is used when intermediate estimates are shown to be close to their targets with high probability (e.g., sample average convergence via Chebyshev).
- L^2 **convergence** (mean square) appears in the linear TD convergence analysis with function approximation.
- **Borel–Cantelli** is the standard technique for upgrading convergence in probability to almost sure convergence when tail probability sums are finite.

Prerequisites. Lesson 5 (random variables, expectation), Lesson 7 (variance, independence), Lesson 9 (Markov and Chebyshev inequalities).

2 References

The following sources complement this lesson:

- Durrett [2019], Chapter 2 (Sections 2.1–2.3): modes of convergence and their relationships.
- Billingsley [1995], Chapter 5 (Sections 25–27): convergence of measurable functions and the Borel–Cantelli lemmas.
- Williams [1991], Chapters 4–5: almost sure convergence, convergence in probability, and the role of Borel–Cantelli in martingale theory.
- Shiryaev [2016], Chapter II, Sections 6–7: convergence modes with detailed examples.

3 Modes of Convergence

Throughout this section, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $\{X_n\}_{n\geq 1}$, X are real-valued random variables defined on it.

3.1 Definitions

Definition 3.1 (Almost Sure Convergence)**.** A sequence $\{X_n\}$ *converges almost surely* (a.s.) to X, written $X_n \xrightarrow{\text{a.s.}} X$, if

$$\mathbb{P}\Big(\Big\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\Big\}\Big) = 1. \tag{1}$$

Equivalently, $\mathbb{P}(\lim_{n\to\infty} X_n = X) = 1$.

Definition 3.2 (Convergence in Probability)**.** $\{X_n\}$ *converges in probability* to X, written $X_n \xrightarrow{\mathbb{P}} X$, if for every $\varepsilon > 0$:

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0. \tag{2}$$

Definition 3.3 (Convergence in L^p)**.** For $p \geq 1$, $\{X_n\}$ *converges in* L^p to X, written $X_n \xrightarrow{L^p} X$, if

$$\lim_{n\to\infty} \mathbb{E}[|X_n - X|^p] = 0. \tag{3}$$

The case $p = 2$ is called *convergence in mean square.*

Definition 3.4 (Convergence in Distribution)**.** $\{X_n\}$ *converges in distribution* to X, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x) \tag{4}$$

at every point x where F_X is continuous.

Remark 3.5 (Convergence in Distribution is Weaker)*.* Convergence in distribution concerns only the distributions (CDFs) of X_n and X, not the joint behavior of X_n and X on the same probability space. In particular, $X_n \xrightarrow{d} X$ does not require $X_n - X$ to be small in any pointwise sense.

Definition 3.6 (Limit Superior of Events)**.** For a sequence of events $\{A_n\}$, the *limit superior* (events occurring *infinitely often*) is

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \{\omega : \omega \in A_n \text{ for infinitely many } n\}. \tag{5}$$

We write $\{A_n \text{ i.o.}\}$ for this event.

4 Hierarchy of Convergence Modes

Convergence Implications

Theorem 4.1 (Hierarchy of Convergence Modes)**.** *The following implications hold:*

$$X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X, \tag{6}$$

$$X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X. \tag{7}$$

No other implications hold in general between these four modes. In particular, almost sure convergence and L^p convergence are incomparable.

We prove each implication and each failure separately.

4.1 L^p Convergence Implies Convergence in Probability

Theorem 4.2 ($L^p \Rightarrow$ Probability)**.** *If $X_n \xrightarrow{L^p} X$ for some $p \geq 1$, then $X_n \xrightarrow{\mathbb{P}} X$.*

Proof. By Markov's inequality (Lesson 9, Theorem 2.1), for any $\varepsilon > 0$:

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(|X_n - X|^p > \varepsilon^p) \leq \frac{\mathbb{E}[|X_n - X|^p]}{\varepsilon^p} \to 0 \tag{8}$$

as $n \to \infty$, since $\mathbb{E}[|X_n - X|^p] \to 0$ by hypothesis. □

4.2 Almost Sure Convergence Implies Convergence in Probability

Theorem 4.3 (a.s. $\Rightarrow$ Probability)**.** *If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{\mathbb{P}} X$.*

Proof. Fix $\varepsilon > 0$. Define $A_n = \{|X_n - X| > \varepsilon\}$ and $B_N = \bigcup_{n=N}^{\infty} A_n$. The sets B_N are decreasing: $B_1 \supseteq B_2 \supseteq \cdots$, and

$$\bigcap_{N=1}^{\infty} B_N = \{\omega : |X_n(\omega) - X(\omega)| > \varepsilon \text{ for infinitely many } n\}. \tag{9}$$

If $X_n(\omega) \to X(\omega)$, then eventually $|X_n(\omega) - X(\omega)| \le \varepsilon$, so $\omega \notin \bigcap_N B_N$. Since almost sure convergence means $\mathbb{P}(\lim X_n = X) = 1$, we get $\mathbb{P}(\bigcap_N B_N) = 0$.

By the continuity of probability from above (Lesson 4, Theorem 3.1):

$$0 = \mathbb{P}\left(\bigcap_{N=1}^{\infty} B_N\right) = \lim_{N\to\infty} \mathbb{P}(B_N) \ge \lim_{N\to\infty} \mathbb{P}(A_N). \tag{10}$$

Since $\mathbb{P}(A_N) \ge 0$, we conclude $\mathbb{P}(A_N) \to 0$, which is $X_n \xrightarrow{\mathbb{P}} X$. □

4.3 Convergence in Probability Implies Convergence in Distribution

Theorem 4.4 (Probability $\Rightarrow$ Distribution)**.** *If $X_n \xrightarrow{\mathbb{P}} X$, then $X_n \xrightarrow{d} X$.*

Proof. Fix x where F_X is continuous. For any $\varepsilon > 0$:

$$\begin{aligned} F_{X_n}(x) &= \mathbb{P}(X_n \le x) \\ &= \mathbb{P}(X_n \le x,\, X \le x+\varepsilon) + \mathbb{P}(X_n \le x,\, X > x+\varepsilon) \\ &\le \mathbb{P}(X \le x+\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon) \\ &= F_X(x+\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon). \end{aligned} \tag{11}$$

For the lower bound, note that $\{X \le x - \varepsilon\} \subseteq \{X_n \le x\} \cup \{|X_n - X| > \varepsilon\}$, so

$$F_{X_n}(x) \ge F_X(x-\varepsilon) - \mathbb{P}(|X_n - X| > \varepsilon). \tag{12}$$

Taking $n \to \infty$ and using $X_n \xrightarrow{\mathbb{P}} X$:

$$F_X(x-\varepsilon) \le \liminf_n F_{X_n}(x) \le \limsup_n F_{X_n}(x) \le F_X(x+\varepsilon). \tag{13}$$

Since F_X is continuous at x, letting $\varepsilon \to 0$ gives $\lim_n F_{X_n}(x) = F_X(x)$. □

4.4 Almost Sure and L^p Convergence Are Incomparable

Theorem 4.5 (a.s. and L^p Are Incomparable)**.** *Neither almost sure convergence implies L^p convergence, nor L^p convergence implies almost sure convergence (for any $p \ge 1$).*

Proof. **a.s. $\not\Rightarrow$ L^p:** Let $\Omega = [0,1]$ with Lebesgue measure, and define $X_n = n \cdot \mathbf{1}_{(0,1/n)}$. For any $\omega > 0$, eventually $1/n < \omega$, so $X_n(\omega) = 0$. Thus $X_n \xrightarrow{\text{a.s.}} 0$. But $\mathbb{E}[|X_n|^p] = n^p \cdot (1/n) = n^{p-1}$, which diverges for $p \ge 1$. In particular, $\mathbb{E}[|X_n|] = 1 \not\to 0$, so $X_n \not\xrightarrow{L^1} 0$.

L^p $\not\Rightarrow$ a.s.: On $\Omega = [0,1]$ with Lebesgue measure, define the "typewriter sequence": enumerate the intervals $[j/k, (j+1)/k]$ for $k = 1, 2, \ldots$ and $j = 0, 1, \ldots, k-1$, and set $X_n = \mathbf{1}_{[j/k,(j+1)/k]}$ where n corresponds to the pair (k, j) via $n = k(k-1)/2 + j + 1$. Then $\mathbb{E}[|X_n|^p] = 1/k \to 0$, so $X_n \xrightarrow{L^p} 0$. But for every $\omega \in [0,1]$, $X_n(\omega) = 1$ for infinitely many n (every subinterval of length $1/k$ eventually covers ω), so $X_n \not\xrightarrow{\text{a.s.}} 0$. □

Remark 4.6 (Converse of Probability $\Rightarrow$ a.s. Fails). The typewriter sequence also shows that convergence in probability does not imply almost sure convergence, since L^p convergence implies convergence in probability (Theorem 4.2), yet $X_n \not\xrightarrow{\text{a.s.}} 0$.

Convergence Modes in TD Learning

The TD(0) convergence theorem (as proved via the Robbins–Siegmund theorem) yields $V_t(s) \xrightarrow{\text{a.s.}} V^\pi(s)$. This is strictly stronger than the convergence in mean square (L^2) that appears in the linear function approximation setting. Understanding that a.s. and L^2 are incomparable explains why separate proof strategies are needed for the tabular and linear-approximation cases.

5 Borel–Cantelli Lemmas

The Borel–Cantelli lemmas connect the summability of event probabilities to almost sure occurrence, forming the primary tool for proving or disproving almost sure convergence.

5.1 First Borel–Cantelli Lemma

First Borel–Cantelli Lemma

Theorem 5.1 (First Borel–Cantelli Lemma). *If $\sum_{n=1}^\infty \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$.*

Proof. Let $B_N = \bigcup_{n=N}^\infty A_n$. By the union bound (Lesson 4, Proposition 3.4):

$$\mathbb{P}(B_N) \leq \sum_{n=N}^\infty \mathbb{P}(A_n). \tag{14}$$

Since $\sum_{n=1}^\infty \mathbb{P}(A_n) < \infty$, the tail sum $\sum_{n=N}^\infty \mathbb{P}(A_n) \to 0$ as $N \to \infty$.

The sets B_N are decreasing: $B_1 \supseteq B_2 \supseteq \cdots$, and $\{A_n \text{ i.o.}\} = \bigcap_N B_N$. By continuity of probability from above:

$$\mathbb{P}(A_n \text{ i.o.}) = \lim_{N\to\infty} \mathbb{P}(B_N) \leq \lim_{N\to\infty} \sum_{n=N}^\infty \mathbb{P}(A_n) = 0. \tag{15}$$

□

Remark 5.2 (No Independence Required). The first Borel–Cantelli lemma makes no assumption about independence. It is a purely measure-theoretic result: summable probabilities imply that only finitely many of the events A_n occur almost surely.

5.2 Second Borel–Cantelli Lemma

Second Borel–Cantelli Lemma

Theorem 5.3 (Second Borel–Cantelli Lemma). *If $\sum_{n=1}^\infty \mathbb{P}(A_n) = \infty$ and $A_1, A_2, \ldots$ are mutually independent, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

Proof. It suffices to show $\mathbb{P}(\bigcup_{n=N}^\infty A_n) = 1$ for every N, since $\{A_n \text{ i.o.}\} = \bigcap_N \bigcup_{n\geq N} A_n$.

Fix N and any $M > N$. By mutual independence, the complements $A_N^c, A_{N+1}^c, \ldots$ are also independent:

$$\mathbb{P}\left(\bigcap_{n=N}^M A_n^c\right) = \prod_{n=N}^M \mathbb{P}(A_n^c) = \prod_{n=N}^M (1 - \mathbb{P}(A_n)). \tag{16}$$

Using the elementary inequality $1 - x \leq e^{-x}$ for $x \geq 0$:

$$\prod_{n=N}^M (1 - \mathbb{P}(A_n)) \leq \exp\left(-\sum_{n=N}^M \mathbb{P}(A_n)\right). \tag{17}$$

Since $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, the partial sums $\sum_{n=N}^{M} \mathbb{P}(A_n) \to \infty$ as $M \to \infty$. Therefore

$$\mathbb{P}\left(\bigcap_{n=N}^{\infty} A_n^c\right) = \lim_{M\to\infty} \mathbb{P}\left(\bigcap_{n=N}^{M} A_n^c\right) = 0, \tag{18}$$

where the limit exchange uses continuity from above applied to the decreasing sequence $\bigcap_{n=N}^{M} A_n^c$. Taking complements: $\mathbb{P}(\bigcup_{n=N}^{\infty} A_n) = 1$. □

Example 5.4 (Monkey Typing Shakespeare)**.** A monkey types random letters uniformly on a 26-letter alphabet. Let A_n be the event that the n-th through $(n+5)$-th letters spell "HAMLET." Then $\mathbb{P}(A_n) = (1/26)^6 > 0$ for each n. Since $\sum_n \mathbb{P}(A_n) = \infty$ and the events corresponding to non-overlapping blocks are independent, a modification of the second Borel–Cantelli lemma (applied to the non-overlapping subsequence) shows that $\mathbb{P}$("HAMLET" appears i.o.) $= 1$.

6 Borel–Cantelli Characterization of Almost Sure Convergence

Theorem 6.1 (Borel–Cantelli Characterization)**.** *A sequence $\{X_n\}$ converges almost surely to X if and only if for every $\varepsilon > 0$:*

$$\mathbb{P}(|X_n - X| > \varepsilon \text{ i.o.}) = 0. \tag{19}$$

Proof. (⇒) Suppose $X_n \xrightarrow{\text{a.s.}} X$. For each fixed $\varepsilon > 0$, the set of ω where $|X_n(\omega) - X(\omega)| > \varepsilon$ for infinitely many n is contained in the set where $X_n(\omega) \not\to X(\omega)$. The latter has probability 0, so $\mathbb{P}(|X_n - X| > \varepsilon \text{ i.o.}) = 0$.

(⇐) Suppose (19) holds for every $\varepsilon > 0$. The event "$X_n \not\to X$" equals

$$\{\omega : X_n(\omega) \not\to X(\omega)\} = \bigcup_{k=1}^{\infty} \{|X_n - X| > 1/k \text{ i.o.}\}. \tag{20}$$

By the union bound:

$$\mathbb{P}(X_n \not\to X) \le \sum_{k=1}^{\infty} \mathbb{P}(|X_n - X| > 1/k \text{ i.o.}) = 0. \tag{21}$$

Therefore $\mathbb{P}(X_n \to X) = 1$, i.e., $X_n \xrightarrow{\text{a.s.}} X$. □

Corollary 6.2 (Sufficient Condition for Almost Sure Convergence)**.** *If for every $\varepsilon > 0$, $\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \varepsilon) < \infty$, then $X_n \xrightarrow{a.s.} X$.*

Proof. By the first Borel–Cantelli lemma (Theorem 5.1), the summability condition gives $\mathbb{P}(|X_n - X| > \varepsilon \text{ i.o.}) = 0$ for every $\varepsilon > 0$. By Theorem 6.1, $X_n \xrightarrow{\text{a.s.}} X$. □

Remark 6.3. Corollary 6.2 is the most common method for establishing almost sure convergence in practice. The typical strategy is:

1. Bound $\mathbb{P}(|X_n - X| > \varepsilon)$ using a concentration inequality (Chebyshev, Hoeffding, etc.).
2. Verify that the resulting bound is summable over n.
3. Apply the corollary.

Borel–Cantelli in Stochastic Approximation

In the convergence proof of stochastic approximation algorithms (Robbins–Monro), one often shows that $\mathbb{P}(|\theta_n - \theta^*| > \varepsilon) \le C/n^{1+\delta}$ for some $\delta > 0$, where θ_n is the iterate and θ^* is the fixed point. Since $\sum_n n^{-(1+\delta)} < \infty$, Corollary 6.2 upgrades this to $\theta_n \xrightarrow{\text{a.s.}} \theta^*$. The

same technique applies to TD learning iterates $V_t(s)$ when convergence rates are available from Lyapunov analysis.

7 Exercises

Exercise 7.1 ((*)). Let $X_n \xrightarrow{d} c$ where c is a constant (i.e., $F_{X_n}(x) \to \mathbf{1}_{\{x \geq c\}}$ at continuity points). Prove that $X_n \xrightarrow{\mathbb{P}} c$. (Thus, for convergence to a constant, convergence in distribution and convergence in probability are equivalent.)

Exercise 7.2 ((*)). Let $X_n \xrightarrow{\text{a.s.}} X$ and suppose $|X_n| \leq M$ a.s. for all n and some constant $M > 0$. Prove that $X_n \xrightarrow{L^p} X$ for every $p \geq 1$.
Hint: Apply the Dominated Convergence Theorem (Lesson 11, Theorem 4.5) to $|X_n - X|^p$.

Exercise 7.3 ((**)). Prove that if $X_n \xrightarrow{\mathbb{P}} X$, then there exists a subsequence $\{X_{n_k}\}$ such that $X_{n_k} \xrightarrow{\text{a.s.}} X$.
Hint: Choose n_k so that $\mathbb{P}(|X_{n_k} - X| > 1/k) < 2^{-k}$, then apply the first Borel–Cantelli lemma.

Exercise 7.4 ((**)). Let A_n be events (not necessarily independent) with $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. Construct an explicit example where $\mathbb{P}(A_n \text{ i.o.}) = 0$, showing that independence is essential in the second Borel–Cantelli lemma.
Hint: Consider $A_n = A$ for all n, with $\mathbb{P}(A) = 0$. Find a less trivial example where each $\mathbb{P}(A_n) > 0$.

Exercise 7.5 ((**)). Let $X_n = n^{1/2}\mathbf{1}_{(0,1/n)}$ on $(\Omega, \mathcal{F}, \mathbb{P}) = ([0,1], \mathcal{B}([0,1]), \lambda)$, where λ is the Lebesgue measure. Determine which of the four modes of convergence hold for $X_n \to 0$. Justify each answer.

Exercise 7.6 ((***)). **(Converse Borel–Cantelli for pairwise independent events.)** Let $A_1, A_2, \ldots$ be pairwise independent events with $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. Prove that $\mathbb{P}(A_n \text{ i.o.}) = 1$.
Hint: Compute $\mathrm{Var}(S_N)$ where $S_N = \sum_{n=1}^{N} \mathbf{1}_{A_n}$, using pairwise independence to bound the variance. Then apply Chebyshev's inequality to show $\mathbb{P}(S_N > 0) \to 1$.

Exercise 7.7 ((***)). Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_1] = \mu$ and $\mathbb{E}[X_1^4] < \infty$. Using Markov's inequality with the fourth moment and Corollary 6.2, prove that $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\text{a.s.}} \mu$.
Hint: Show $\mathbb{E}[(\bar{X}_n - \mu)^4] = O(1/n^2)$, then conclude $\sum_n \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) < \infty$.

8 Summary

- **Four modes of convergence:** almost sure ($X_n \xrightarrow{\text{a.s.}} X$), in probability ($X_n \xrightarrow{\mathbb{P}} X$), in L^p ($X_n \xrightarrow{L^p} X$), and in distribution ($X_n \xrightarrow{d} X$).
- **Hierarchy:** a.s. $\Rightarrow$ prob $\Rightarrow$ dist and $L^p \Rightarrow$ prob $\Rightarrow$ dist. No other general implications hold; a.s. and L^p are incomparable.
- **First Borel–Cantelli:** $\sum \mathbb{P}(A_n) < \infty \Rightarrow \mathbb{P}(A_n \text{ i.o.}) = 0$. No independence needed.
- **Second Borel–Cantelli:** $\sum \mathbb{P}(A_n) = \infty$ plus independence $\Rightarrow \mathbb{P}(A_n \text{ i.o.}) = 1$.
- **Characterization:** $X_n \xrightarrow{\text{a.s.}} X$ iff $\mathbb{P}(|X_n - X| > \varepsilon \text{ i.o.}) = 0$ for all $\varepsilon > 0$.
- **Sufficient condition:** $\sum_n \mathbb{P}(|X_n - X| > \varepsilon) < \infty$ for all $\varepsilon > 0$ implies $X_n \xrightarrow{\text{a.s.}} X$.
- **Strategy:** bound tail probabilities via concentration inequalities, verify summability, apply Borel–Cantelli.

9 Further Reading

- **Lesson 9** develops the probability inequalities (Markov, Chebyshev, Hoeffding) used to bound tail probabilities in the Borel–Cantelli strategy.
- **Lesson 11** proves the limit theorems (WLLN, SLLN, CLT) that rely on the convergence modes and Borel–Cantelli lemmas established here.
- **Lesson 13** on martingale convergence provides the supermartingale convergence theorem, which gives almost sure convergence directly without Borel–Cantelli.
- Durrett [2019], Chapter 2 gives complete proofs of all convergence implications with additional counterexamples.
- Billingsley [1995], Section 27 provides the definitive treatment of convergence in distribution, including the portmanteau theorem characterizing equivalent conditions.
- Grimmett and Stirzaker [2001], Chapter 7 covers convergence of random variables with many worked examples.

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